

LITCHFIELD, IL, USA

WMO: 722972

Lat: 39.163N

Lon: 89.675W

Elev: 690

StdP: 14.33

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.3	9.1	-5.3	4.3	5.6	-0.1	5.6	9.7	28.9	29.6	26.8	31.7	7.3	310	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	93.3	74.6	90.5	74.7	88.3	73.9	78.8	87.9	77.4	86.7	75.9	84.9	8.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	141.9	84.8	74.6	132.8	82.8	73.0	125.8	81.0	42.9	88.3	41.2	87.0	39.8	84.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.3	(b) 21.9	(c) 19.2	(e) -2.7	(f) 95.9	(g) 6.6	(h) 4.2	(i) -7.5	(j) 98.9	(k) -11.3	(l) 101.4	(m) -15.1	(n) 103.8	(o) -19.9	(p) 106.9		
(a) 25.3	(b) 21.9	(c) 19.2	(e) -3.6	(f) 81.6	(g) 6.4	(h) 1.5	(i) -8.1	(j) 82.6	(k) -11.9	(l) 83.5	(m) -15.4	(n) 84.3	(o) -20.0	(p) 85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	29.4	32.0	43.9	54.5	64.5	73.6	75.8	74.7	68.3	56.0	43.5	33.1
	DBStd	18.96	11.41	11.48	11.23	9.27	8.71	5.25	5.59	5.61	7.33	9.63	9.68	9.57
	HDD50	2285	643	511	258	59	5	0	0	0	0	45	236	528
	HDD65	5135	1105	924	659	326	120	4	2	2	49	311	644	989
	CDD50	3830	3	7	67	195	455	708	801	767	549	232	42	4
	CDD65	1205	0	0	4	12	105	262	337	304	148	33	0	0
	CDH74	10980	0	0	19	133	835	2289	3324	2770	1331	275	4	0
	CDH80	4096	0	0	1	18	226	827	1358	1103	491	72	0	0
Wind	WSAvg	8.6	10.8	10.1	10.4	10.5	8.8	7.2	5.6	5.2	6.0	8.4	9.8	10.0
Precipitation	PrecAvg	39.2	2.6	2.0	2.8	4.0	5.2	3.8	3.3	3.2	3.0	3.1	3.7	2.3
	PrecMax	51.0	7.8	4.4	5.5	11.0	11.5	7.8	6.1	5.7	8.9	6.5	7.3	7.2
	PrecMin	32.2	0.5	0.5	0.6	0.9	1.3	1.8	1.2	1.5	0.2	1.1	0.5	0.8
	PrecStd	5.6	1.7	1.2	1.3	2.4	3.1	1.7	1.4	1.2	2.0	1.4	1.8	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.4	65.3	76.9	81.9	90.1	92.8	99.7	98.6	93.1	87.2	72.7	62.0
		MCWB	56.4	57.3	63.6	63.9	71.0	74.2	74.8	72.8	71.1	71.2	60.9	58.2
	2%	DB	55.3	59.8	71.8	78.5	85.6	90.4	93.1	93.4	89.6	81.3	66.3	55.5
		MCWB	52.0	54.3	61.6	63.6	70.9	74.4	76.1	73.8	72.0	66.8	58.6	51.5
	5%	DB	50.5	54.4	65.9	73.4	82.1	88.2	90.2	89.9	86.2	76.4	62.7	51.2
		MCWB	47.1	49.7	58.5	60.9	68.7	73.8	76.4	74.5	71.2	63.8	55.9	48.1
	10%	DB	44.8	48.2	61.2	70.1	79.0	85.2	87.6	86.2	81.7	71.8	58.5	46.4
		MCWB	41.7	44.3	54.9	59.9	67.4	72.1	75.4	73.6	68.0	61.8	52.6	43.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.5	59.5	66.1	68.6	75.1	79.0	81.0	80.9	77.3	73.0	64.7	58.8
		MCDB	60.7	63.3	73.8	78.7	84.9	88.0	90.0	89.0	86.2	83.4	69.0	61.3
	2%	WB	53.3	55.6	62.3	65.8	72.8	77.1	79.3	78.3	74.9	70.2	60.3	53.3
		MCDB	55.3	59.5	70.3	73.7	82.0	86.9	88.7	87.2	84.4	77.9	65.1	54.9
	5%	WB	47.2	50.3	59.3	63.5	71.1	75.4	77.9	76.6	72.9	67.3	56.9	48.1
		MCDB	49.6	54.2	64.5	70.8	79.5	84.7	87.1	85.7	81.6	72.2	61.4	50.6
	10%	WB	41.3	44.6	55.9	60.9	69.0	73.8	76.4	74.9	71.1	63.9	53.2	44.0
		MCDB	44.9	48.3	60.6	68.1	76.7	82.7	85.2	83.3	78.7	69.4	57.9	46.5
Mean Daily Temperature Range	5% DB	MDBR	14.5	15.4	17.1	19.6	18.7	18.7	19.2	19.7	21.7	20.5	17.0	13.0
		MCDBR	21.9	23.5	22.6	24.4	21.6	21.5	21.7	23.2	25.0	24.9	22.8	19.3
		MCWBR	18.7	18.6	14.7	13.8	10.1	9.4	9.1	9.1	10.2	12.3	15.8	16.1
	5% WB	MCDBR	20.8	22.3	20.3	21.0	19.0	19.1	19.1	19.5	20.4	20.7	19.9	17.8
		MCWBR	19.0	18.8	14.2	14.2	10.2	9.4	9.4	9.7	9.8	11.8	15.9	16.1
Clear-Sky Solar Irradiance	taub	0.299	0.310	0.344	0.392	0.428	0.428	0.462	0.449	0.405	0.351	0.306	0.295	
	taud	2.469	2.429	2.404	2.269	2.224	2.260	2.157	2.184	2.289	2.449	2.518	2.497	
	Ebn at Noon	279	290	290	280	271	271	260	260	265	271	274	272	
	Edh at Noon	25	30	34	42	44	43	47	45	38	29	24	23	
All-Sky Solar Radiation	RadAvg	620	870	1174	1555	1751	2014	1984	1769	1496	1058	712	520	
	RadStd	70	85	97	93	147	149	132	108	100	116	72	61	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+148

Nomenclature: See separate page